

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

How Agents Think

Roadmap to reasoning models

1

LLMs and temperature

How next-token prediction works, and how temperature changes it

- P(next | previous)
- Temperature

2

Slow vs. fast reasoning

A dual-process view of what we want LLMs to do

- System 1
- System 2

3

Prompt engineering

Getting better reasoning through the prompt

- Chain-of-thought

4

Beyond prompt engineering

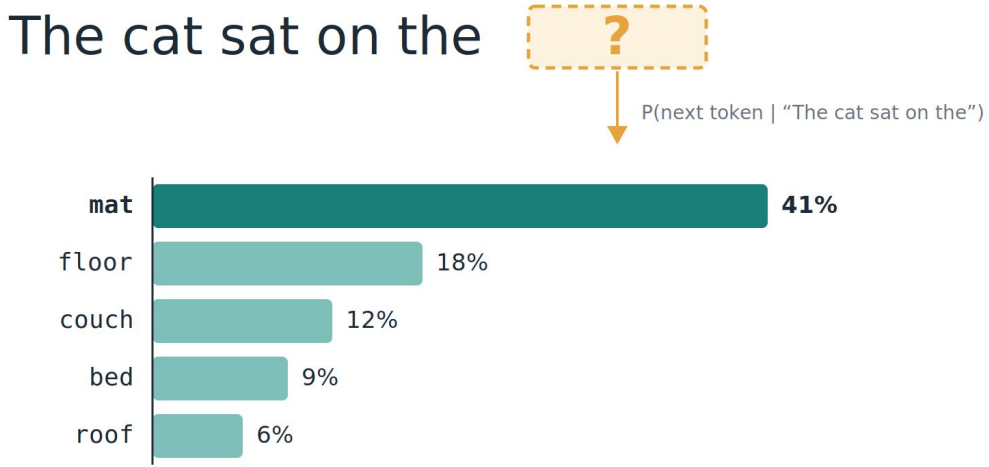
Searching over reasoning paths

- Self-consistency
- Tree of Thoughts
- Graph of Thoughts

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Next-token prediction

- LLM models: P(next token | previous tokens)
- Autoregressive generation: sample a token, append it, repeat
- The model "thinks" only by writing tokens

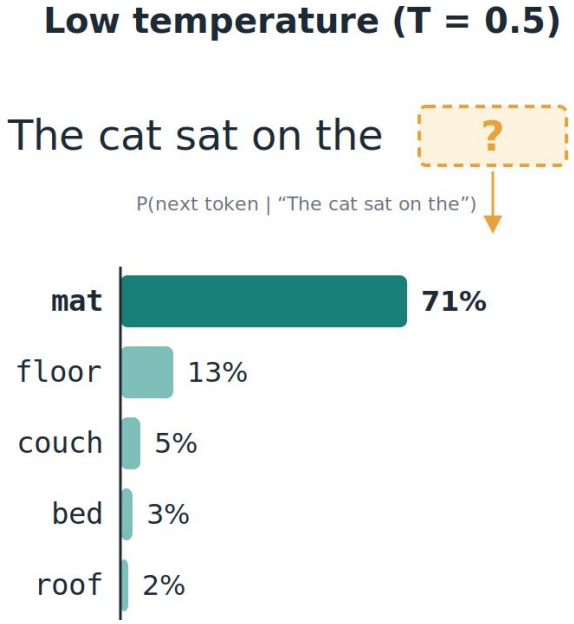


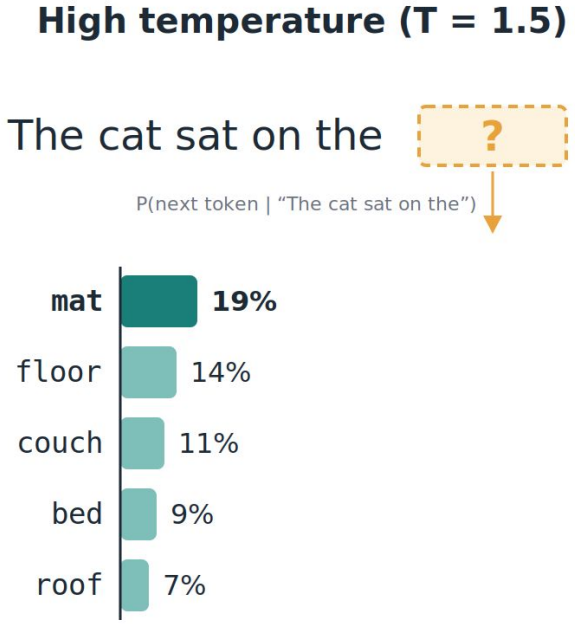
THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Temperature

- **Temperature (T)** rescales probabilities
  - High T (>1):
    - Probability distribution closer to uniform
    - More creative, more error prone
  - Low T (<1):
    - Closer to deterministic behavior
    - More reliable and reproducible
  - T=0
    - Next token is the one with maximum probability
    - Closest to deterministic, but provider side randomness remains

Temperature





THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Temperature

- Controlling temperature:

```
from openai import OpenAI

client = OpenAI()

response = client.responses.create(
     model='beam-1.0-bf16',
     input='How do you show that pi is irrational?',
     temperature=0.9,
)

print(response.output_text)
```


THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Dual Process Theory

System 1

Fast, automatic, intuitive, subconscious

Example tasks

autocompletion, spam classification



System 2

Slow, deliberate, step-by-step, logical, conscious

Example tasks

debugging code, solving math problems

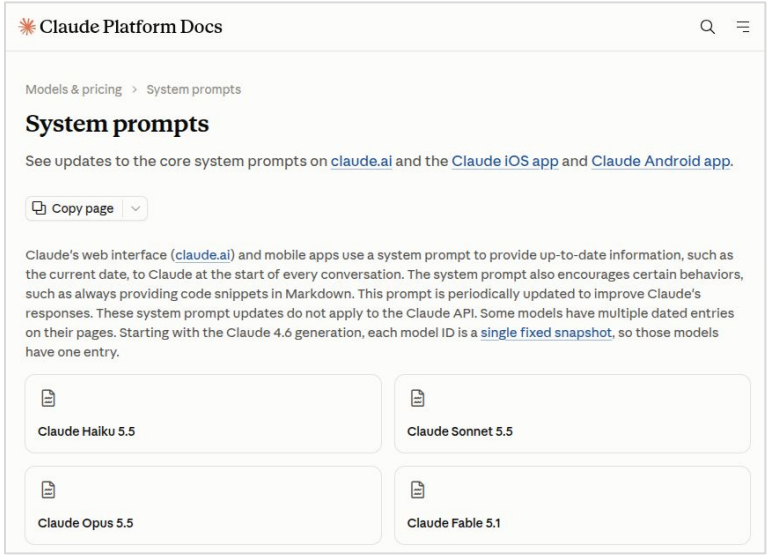
- Originated in philosophy and psychology
- Popularized by Daniel Kahneman, '*Thinking, Fast and Slow*' [1]

Reasoning techniques aim to push the model toward System 2: spend more tokens and compute before answering

Levers for better reasoning

1	**Better instructions**	System prompt, Few-shot prompting
2	**Elicit intermediate steps**	Chain-of-thought
3	**Sample and aggregate**	Self-consistency
4	**Search over reasoning paths**	Tree / Graph of Thoughts
5	**Constrained output**	Schemas, JSON, Pydantic

System prompt



Source: [9] Anthropic, System prompts (release notes). https://platform.claude.com/docs/en/release-notes/system-prompts/overview

- Separate from user prompt
- Sets highest-level behavior: role, goals, constraints, tone, output format
- Persists across the conversation
- Commercial models have an internal one, typically not publicly available
  - Anthropic is a notable exception

System prompt: Claude Sonnet 3.5

Excerpts from Claude Sonnet 3.5 (2024)

```
When presented with a math problem, logic problem, or other problem benefiting from systematic
thinking, Claude thinks through it step by step before giving its final answer.
```

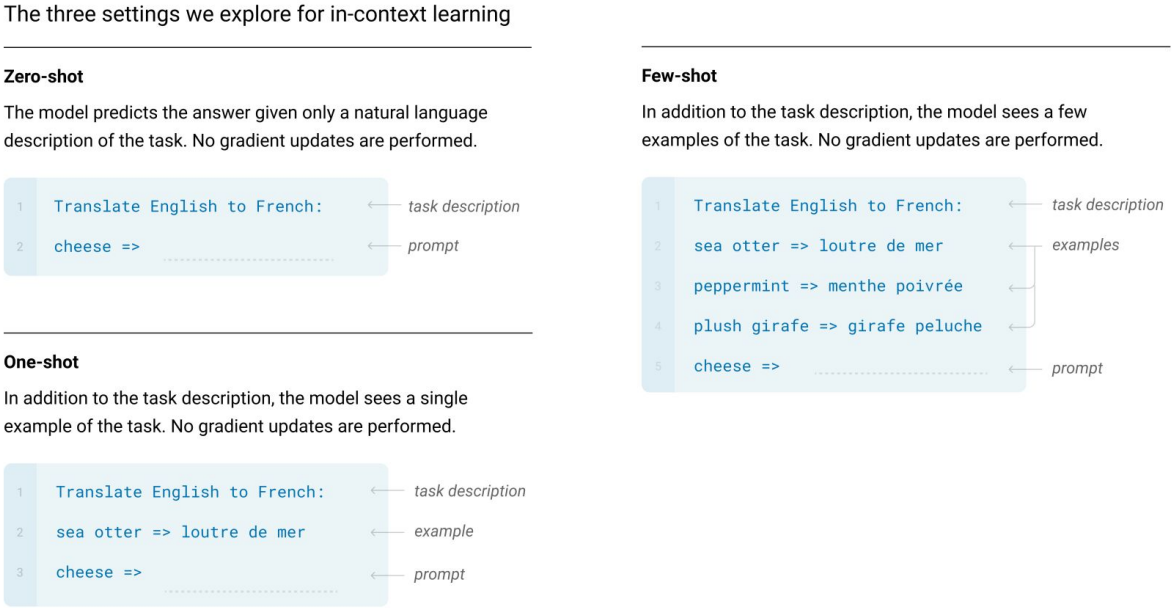
```
If Claude believes the human is asking for something harmful, it doesn't help with the harmful thing.
Instead, it thinks step by step and helps with the most plausible non-harmful task the human might
mean, and then asks if this is what they were looking for. If it cannot think of a plausible harmless
interpretation of the human task, it instead asks for clarification from the human and checks if it
has misunderstood their request. Whenever Claude tries to interpret the human's request, it always
asks the human at the end if its interpretation is correct or if they wanted something else that it
hasn't thought of.
```

Source: [9] Anthropic, Claude Sonnet 3.5 system prompts (claude.ai).
https://platform.claude.com/docs/en/release-notes/system-prompts/claude-sonnet-3-5

Few-shot prompting

- In-context learning
  - Models learn how to solve a task from prompting, without updates in the model weights
  - Emerged as a major capability with foundation models (GPT-3)
- Few-shot prompting:
  - Include k input-output examples in the prompt

Few-shot prompting: example



[2] Brown, T., et al. (2020). *Language Models are Few-Shot Learners.* NeurIPS. arXiv:2005.14165
https://arxiv.org/pdf/2005.14165

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Chain-of-Thought prompting (CoT)

- Modification of few-shot prompting to address logic, math and symbolic reasoning
- Examples in the prompt include worked reasoning, not just final answers
- Demonstrates both the *format* and the *style* of reasoning

Chain-of-Thought (CoT): example

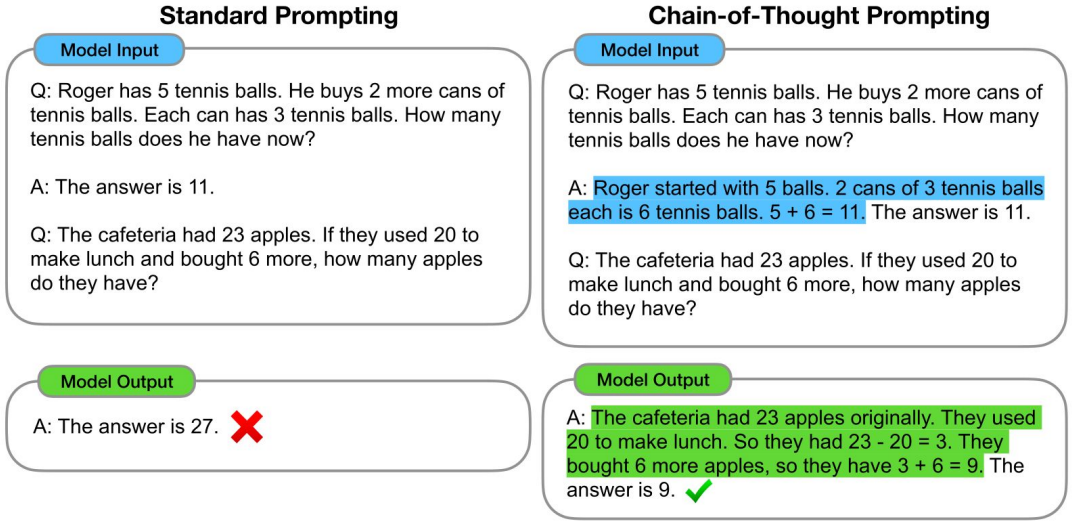


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

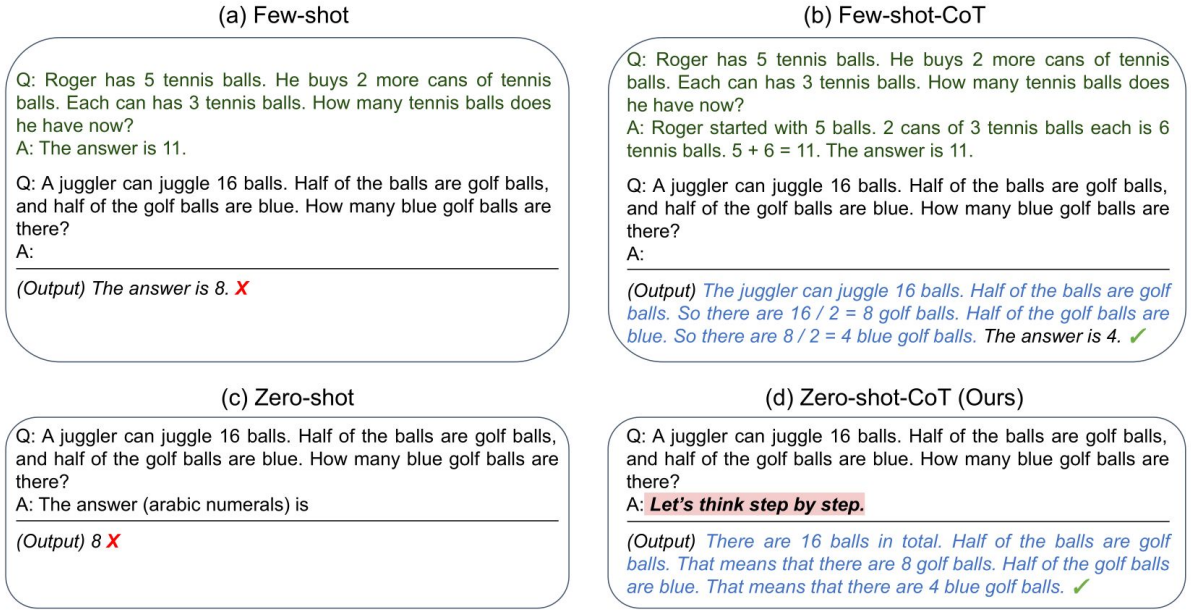
[3] Wei, J., et al. (2022). *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.* NeurIPS. arXiv:2201.11903 https://arxiv.org/pdf/2201.11903

Zero-shot CoT

- Alternative approach to CoT
- Obtained by prompting
  - Trigger phrase that elicits reasoning, e.g. "*Let's think step by step*"
  - Two-stage version: "First generate reasoning, then extract the answer"
- No examples needed
  - Cheap and general-purpose
  - Usually weaker than carefully built few-shot CoT

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Zero-shot CoT: example

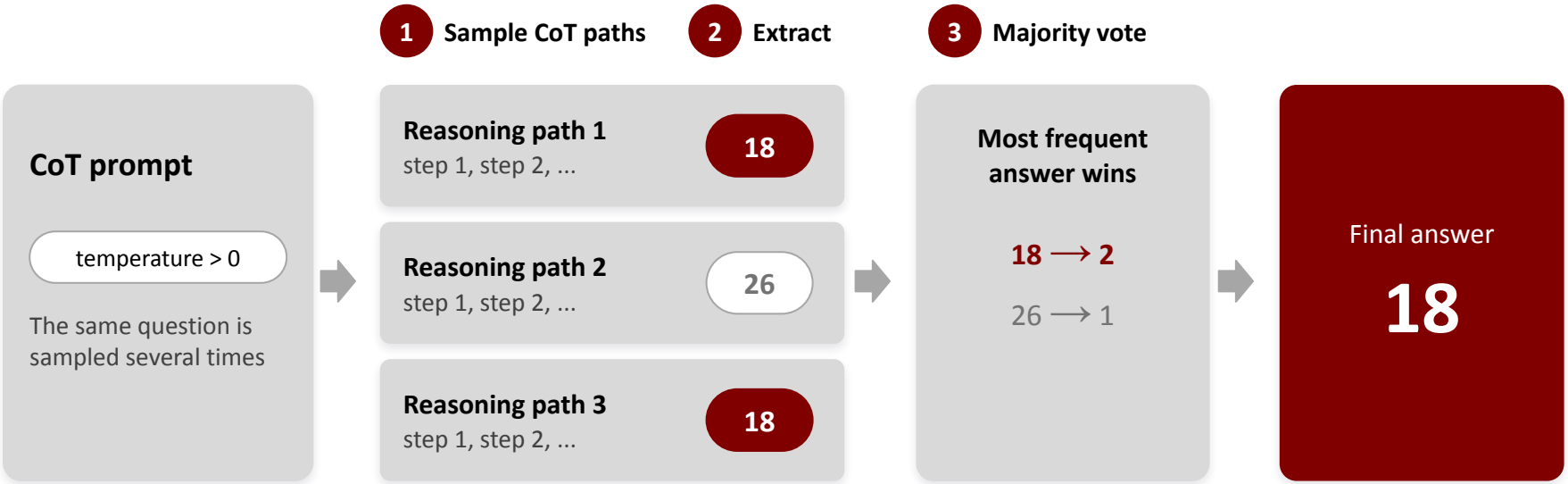


[4] Kojima, T., et al. (2022). *Large Language Models are Zero-Shot Reasoners.* NeurIPS. arXiv:2205.11916
https://arxiv.org/pdf/2205.11916

Self-consistency

- Procedure:
  1. Sample multiple CoT responses at temperature > 0
  2. Extract each final answer
  3. Take a majority vote
- Improves accuracy on math and commonsense benchmarks
- Increased cost in tokens and latency

Self-consistency

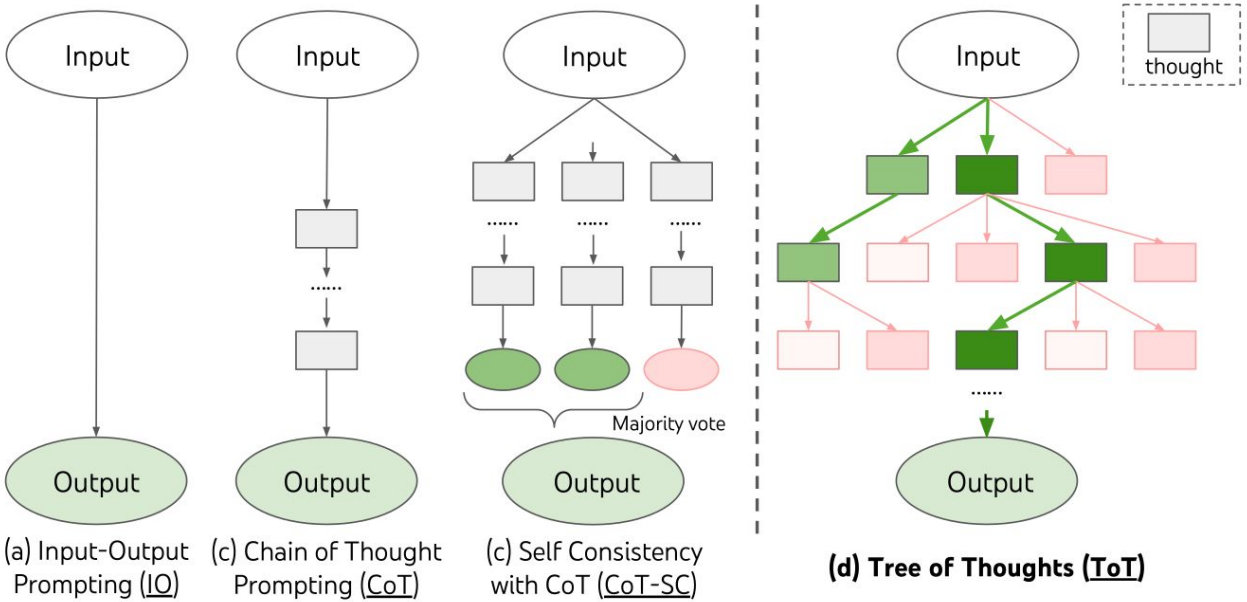


THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Tree of Thoughts (ToT)

- Extension of CoT
- Treats reasoning as **search** over a tree of coherent intermediate steps
- Four components:
  1. Thought decomposition
  2. Thought generation (propose several candidates)
  3. State evaluation (the LLM scores or votes on each)
  4. Search algorithm (with pruning and backtracking)

Tree of Thoughts (ToT)



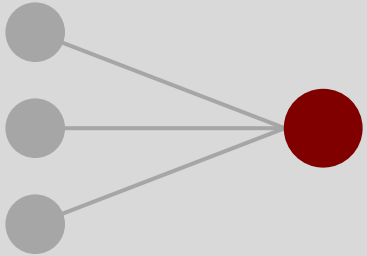
[6] Yao, S., et al. (2023). *Tree of Thoughts: Deliberate Problem Solving with Large Language Models.* NeurIPS. arXiv:2305.10601 https://arxiv.org/pdf/2305.10601

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Graph of Thoughts (GoT)

Generalizes the tree to an arbitrary graph: thoughts are vertices, dependencies are edges

Aggregation



Merge several thoughts into a better one

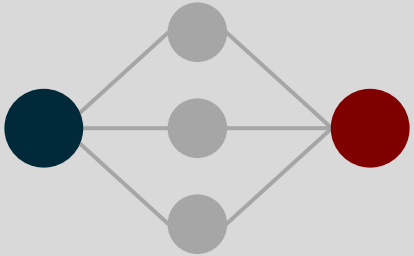
Refinement



repeat until good enough

Loop on a thought to improve it

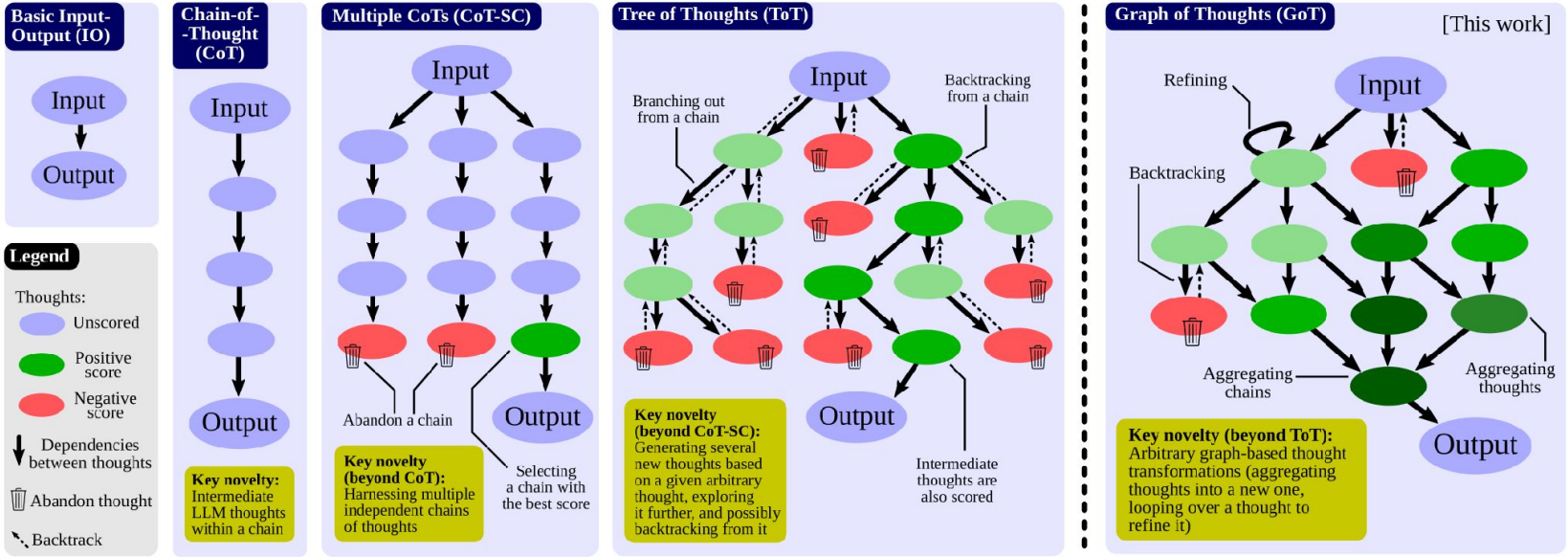
Decomposition



Split into subproblems that are solved and recombined

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Graph of Thoughts (GoT): comparison



[7] Besta, M., et al. (2024). *Graph of Thoughts: Solving Elaborate Problems with Large Language Models.* AAAI. arXiv:2308.09687 https://arxiv.org/pdf/2308.09687

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Activity 2: Self-consistency and Tree-of-thought

1. **Use an LLM to review a paper using two patterns**
   - Self-consistency
   - Tree-of-thought

2. **Reflect**
   - Which of the reviews seem better?
   - How could we evaluate the reviews?
   - How could human input improve the reviews?

Activity 2: Self-consistency and Tree-of-thought

Self-consistency task

- Write code prompting a model to review a paper for publication and producing a score
- Run that model call N times
- Aggregate scores to obtain a final paper score
  - Choose: most frequent, average, …

Activity 2: Self-consistency and Tree-of-thought

Tree-of-thought task

- Run the paper review code N times and collect the complete review for each run
- Prompt a model to evaluate each review and give it a score
- Produce a final score for the paper taking into account each review score and the corresponding evaluation scores

References

[1] **System 1 vs System 2:** Kahneman, D. (2011). *Thinking, Fast and Slow.* Farrar, Straus and Giroux.

[2] **In-context learning:** Brown, T., et al. (2020). *Language Models are Few-Shot Learners.* NeurIPS. arXiv:2005.14165

[3] **CoT:** Wei, J., et al. (2022). *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.* NeurIPS. arXiv:2201.11903

[4] **Zero-shot CoT:** Kojima, T., et al. (2022). *Large Language Models are Zero-Shot Reasoners.* NeurIPS. arXiv:2205.11916

References (cont.)

[5] **Self-consistency:** Wang, X., et al. (2023). *Self-Consistency Improves Chain of Thought Reasoning in Language Models.* ICLR. arXiv:2203.11171

[6] **ToT:** Yao, S., et al. (2023). *Tree of Thoughts: Deliberate Problem Solving with Large Language Models.* NeurIPS. arXiv:2305.10601

[7] **GoT:** Besta, M., et al. (2024). *Graph of Thoughts: Solving Elaborate Problems with Large Language Models.* AAAI. arXiv:2308.09687

[8] **Pydantic documentation:** https://docs.pydantic.dev

[9] **Anthropic system prompts:** https://platform.claude.com/docs/en/release-notes/system-prompts/overview